



Syed Ali Hamza

📍 **Home** : B1 168 F/2 Mohallah Mujahidabad Jhelum, Punjab, Pakistan Postal Code: 49600

✉ **Email**: [syedaliamza108@gmail.com](mailto:syedalihamza108@gmail.com) 📞 **Phone**: (+92) 3175945735

🌐 **LinkedIn**: <https://www.linkedin.com/in/syedalihamza124/>

Passport: DA6212021 **Date of birth**: 10/10/2002 **Nationality**: Pakistani

ABOUT ME

Motivated graduate with a strong background in biological sciences and research methods. Good analytical thinking and problem-solving skills, with a strong interest in scientific innovation. Has experience working both independently and as part of a team in academic settings. Committed to continuous learning and contributing to meaningful research.

WORK EXPERIENCE

SUNDAS FOUNDATION

City: LAHORE | **Country**: Pakistan

[06/09/2024 – 20/09/2024]

INTERNSHIP

- Performed hematological testing, infection control, and patient-focused laboratory practices to support safe transfusions.
- Ensured sample compatibility and collaborated with clinical staff to maintain workflow efficiency and timely reporting.
- Streamlined laboratory processes, helping to reduce patient waiting times by 10% and enhance care delivery.
- Collected, processed, and analyzed blood samples for 60+ patients daily, ensuring high diagnostic accuracy.

EDUCATION AND TRAINING

[15/09/2021 – 20/07/2025]

BACHELOR OF SCIENCE (BS) IN BIOTECHNOLOGY (2025)

UNIVERSITY OF CENTRAL PUNJAB <https://www.ucp.edu.pk/>

City: LAHORE | **Country**: Pakistan | **Field(s) of study**: Natural sciences, mathematics and statistics: • *Biological and related sciences not further defined* | **Final grade**: 3.71 | **Level in EQF**: EQF level 6 | **NQF Level**: 6 | **Thesis**: Exploring Computational Insights into Microbial Gut Dynamics and Neurodegeneration in Parkinsons Disease Using RNA-SEQ Analysis.

- Core Biotechnology: Molecular Biology, Microbiology, Immunology, Nanotechnology, and Bioinformatics.
- Laboratory Techniques: PCR, ELISA, Gel Electrophoresis, Microbial Culture, Nanoparticle Synthesis
- Research & Communication: Biostatistics, Experimental Design, Scientific Writing, Presentations
- Chemical Sciences: Biochemistry, Organic & Analytical Chemistry (spectrophotometry, titration, synthesis)



PROJECTS

[01/10/2024 – 01/07/2025]

Exploring Computational Insights into Microbial Gut Dynamics and Neurodegeneration in Parkinsons Disease Using RNA-SEQ Analysis

- Pioneered a custom bioinformatics pipeline for comparative genomic analysis, uncovering critical transcriptomic differences between *healthy* and *Parkinson's Disease through RNA sequences*.
- Leveraged advanced Python scripting for efficient sequence parsing, high-throughput alignment (BLAST), and robust differential expression analysis.
- Analyzed the gut microbiome (***Porphyromonas gingivalis***, ***Collinsella Aerofaciens***) to study links with neurodegeneration and discovered novel biomarkers.

[10/03/2024 – 10/10/2024]

Exploring Polyurethanase-Driven Bioremediation: Structural Insights, Enzyme Substrate Dynamics and Environmental strategies for urethane degradation – Insilico Research

- Conducted in-silico molecular modeling and dynamics simulations of polyurethanase from ***Serratia proteamaculans*** to evaluate its bioremediation potential for urethane degradation.
- Demonstrated strong enzyme-substrate interactions and structural stability, emphasizing potential applications in sustainable environmental cleanup.
- Developed and presented a scientific poster, achieving 3rd position for research excellence and presentation quality.

PUBLICATIONS

[2026]

Molecular cloning and expression of *Bacillus cereus* xylanase gene in *E. coli* BL21

- Successfully cloned the ***Bacillus cereus* xylanase gene** (~1071 bp) into the pET-21c(+) vector and expressed it in *E. coli* BL21.
- Verified recombinant plasmid (~6512 bp) through colony PCR and **BamHI** digestion.
- Established a scalable platform for thermostable **bacterial xylanase production**, applicable in pulp, food, and biofuel industries.

Authors: Rizvi, A. I., Nawaz, A., Afzal, R., Asim, M., Hamza, S. A., Shahid, T., ... Hanif, N. (2026). Journal of Applied Biological Sciences, 20(1), 34–40

Link: <https://doi.org/10.71336/jabs.1499>

[2025]

Unravelling Gut microbial integrative transcriptomics and host gene expression in Parkinson's disease through computational RNASeq Analysis

- Conducted computational RNA-Seq analysis integrating **gut microbial** and host transcriptomics in Parkinson's disease.
- Identified dysbiosis signatures (***Porphyromonas gingivalis***, ***Collinsella aerofaciens***) linked to barrier dysfunction, immune imbalance, and neuroinflammation.

Authors: Syed Ali Hamza, Muhammad Rehan, Hira Mubeen, Muhammad Asjad Haseeb | **Volume, Issue and Pages:** Preprint paper, Not peer reviewed

Link: <https://www.researchsquare.com/article/rs-7768637/v1>

SKILLS

Technical & Laboratory Skills

PCR | Gel Electrophoresis | DNA&RNA Extraction | ELISA | Spectrophotometry | Microscopy | Nanoparticle Synthesis (Green&Chemical Methods) | Microbial Culture | Antimicrobial Susceptibility Testing



Bioinformatics & Computational Skills

Python(Basic) | Protein Characterization | Genome Alignment | NCBI | BLAST | Swiss Dock | Discovery Studio | ClustalW | AlphaFold | PSIPRED | SnapGene | ADDGene | SRA TOOLKIT

Professional & Research Skills

Scientific Writing | Data Analysis | Project Management | Time Management | Problem-Solving | Collaboration | Adaptability | Research Skills | Written and Verbal communication

CERTIFICATIONS

[Coursera]

Researcher's guide to DNA sequencing data

- Applied bioinformatics tools for sequence analysis and quality evaluation.
- Acquired practical skills in DNA sequencing workflows and data management.
- Enhanced understanding of genomic research methods and data interpretation.

Link: <https://coursera.org/share/dce241fb59fab4feecdeba27689cd015>

[American Society for
Microbiology, 2025]

Global Outreach Student Member- Certificate of Membership

CONFERENCES AND SEMINARS

[16/11/2024]

The Blueprint of Life – Where Biology Meets Technology West Visayas State University, Philippines

[03/07/2025 – 05/07/2025]

Industry Panel: Career Beyond Research (3-day program)

Biversity / Tabs Learning Pvt. Ltd., Chennai, India

LANGUAGE SKILLS

Mother tongue(s): Urdu

Other language(s):

English

LISTENING B2 READING C1 WRITING B2

SPOKEN PRODUCTION B2 SPOKEN INTERACTION B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

VOLUNTEERING

[09/2024]

Services for Humanity Sundas Foundation Lahore, Pakistan

- Donated blood to support patients with thalassemia and hematological conditions.
- Recognized by the Thalassaemia International Federation for humanitarian service.
- Contributed to community health and voluntary service initiatives.